

$$P(A/B) = \frac{P(A \cap B)}{P(B)}, \text{ si } P(B) > 0$$

$$P(A_j/B) = \frac{P(A_j)P(B/A_j)}{\sum_{i=1}^n P(A_i)P(B/A_i)}$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$A \text{ y } B \text{ son sucesos independientes} \leftrightarrow P(A \cap B) = P(A) \cdot P(B)$$

$$\forall x \in R_x : p(x) = P(X = x)$$

$$F(x_k) = \sum_{i=1}^k p(x_i)$$

$$E(X) = \sum_{x_i \in R_X} x_i \cdot P(x_i)$$

$$a) P(X \leq x_i) = F(x_i)$$

$$b) P(X \geq x_i) = 1 - F(x_{i-1})$$

$$c) P(x_i \leq X \leq x_j) = F(x_j) - F(x_{i-1})$$

$$d) P(X = x_i) = F(x_i) - F(x_{i-1})$$

$$CV(X) = \frac{\sigma(X)}{E(X)} 100\%$$

$$V(X) = E(X - E(X))^2$$

$$V(X) = E(X^2) - (E(X))^2$$

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}; k: 0, 1, 2, \dots$$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}; k: 0, 1, 2, \dots, n$$

$$V(X) = np(1-p)$$

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx$$

$$E(g(X)) = \int_{-\infty}^{+\infty} g(x) f(x) dx$$

$$V(X) = \int_{-\infty}^{+\infty} (X - E(X))^2 f(x) dx$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt \text{ para todo } x \in R$$

$$v(X) = \frac{(b-a)^2}{12}$$

$$P(X \leq a) = P(X < a) = F(a)$$

$$P(X \geq a) = P(X > a) = 1 - F(a)$$

$$P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = F(b) - F(a)$$

$$f(x) = \lambda e^{-\lambda x} \text{ si } x \geq 0 \ (\lambda > 0)$$

$$E(X) = \frac{1}{\lambda}$$

$$\sigma(X) = \frac{1}{\lambda}$$

$$P(X > t_0 + \Delta t / X > t_0) = P(X > \Delta t)$$

$$P(Z < z) = \Phi(z)$$

$$\text{F.D. } F(x) = \begin{cases} 0 & \text{si } x < 0 \\ 1 - e^{-\lambda x} & \text{si } x \geq 0 \end{cases}$$

$$X \sim N(\mu; \sigma)$$

$$P_X(x) = \sum_y P(x; y)$$

$$P_Y(y) = \sum_x P(x; y)$$

$$P(x/y) = \frac{P(x,y)}{P_Y(y)}$$

$$\frac{X-\mu}{\sigma} = Z \sim N(0;1)$$

$$X \text{ e } Y \text{ son v.a. independientes si y sólo si } P(x, y) = P_X(x) \cdot P_Y(y) \text{ para todo } (x, y)$$

$$Cov(X, Y) = E(XY) - E(Y)E(X)$$

$$V(X + Y) = V(X) + V(Y) + 2Cov(X; Y)$$

$$Cov(aX; bY) = abCov(X; Y)$$

$$\rho(X; Y) = \frac{Cov(X; Y)}{\sigma_X \sigma_Y}$$

$$V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2abCov(X; Y)$$